

Ruthenium rules for new fuel cells

Rice University scientists have fabricated a durable catalyst for high-performance fuel cells by attaching single ruthenium atoms to graphene.

Catalysts that drive the oxygen reduction reaction in fuel cells are usually made of platinum, which stands up to the acidic nature of the cell's charge-carrying electrolyte. But platinum is expensive. The ruthenium-graphene combination may be a suitable replacement, said chemist James Tour, who developed the material with his colleagues at Rice and in China. In tests, its performance easily matched that of traditional platinum-based alloys and bested iron and nitrogen-doped graphene, another contender.

"Ruthenium is often a highly active catalyst when fixed between arrays of four nitrogen atoms, yet it is one-tenth the cost of traditional platinum," Tour said. "And since we are using single atomic sites rather than small particles, there are no buried atoms that cannot react. All the atoms are available for reaction."

The material showed excellent tolerance against methanol crossover and carbon monoxide poisoning in an acidic medium, both of which degrade the efficiency of fuel cells; such degradation is a persistent problem with traditional platinum fuel cells.

Super-strong metal for next tech frontier

Most microscopic sensors, shaped out of silicon, work well in average temperatures, but lose their strength and ability to conduct electronic signals with even modest amounts of heat. Silicon is also very brittle and prone to break. For these reasons, while silicon has been the heart of MEMS technologies for several generations now, the material is not ideal for future uses such as the Internet of Things.

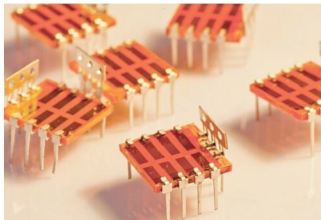
John Hopkins University researchers, in their pursuit of new materials, considered combinations of metals containing nickel, including nickel-based superalloys used to make jet engines. Considering the need for dimensional stability, the researchers experimented with adding the metals molybdenum and tungsten in hopes of curbing the degree pure nickel expands in heat.

In a piece of equipment about the size of a refrigerator in a laboratory at Johns Hopkins, the team hit targets with ions to vaporise the alloys into atoms, depositing them onto a surface, or substrate. This created a film that can be peeled away, thus creating freestanding films with an average thickness of 29 microns—less than the thickness of a human hair.

These freestanding alloy films exhibited extraordinary properties. When pulled, they showed a tensile strength—the ability to maintain shape without deforming or breaking—three times greater than high-strength steel.

Non-toxic alternative for next-generation solar cells

Researchers from the University of Cambridge and the United States have shown that solar cells incorporating bismuth can replicate the properties of lead-based solar cells, but without the same toxicity concerns. Later calculations by another research group showed that bismuth-based cells can convert light into energy at



Bismuth oxyiodide light absorbers (Image courtesy: Steve Penney/University of Cambridge)

efficiencies up to 22 per cent, which is comparable to the most advanced solar cells currently on the market.

Most of the solar cells which we see covering fields and rooftops are made from silicon. Although silicon is highly efficient at converting light into energy, it has a very low 'defect tolerance,' meaning that silicon needs to have very high levels of purity, making it energy-intensive to produce.

For their study, researchers looked at bismuth oxyiodide—a material which was previously investigated for use in solar cells and water splitting, but was not thought to be suitable because of low efficiencies and because it degraded in liquid electrolytes. The researchers used theoretical and experimental methods to revisit this material for possible use in solid-state solar cells.

They found that bismuth oxyiodide is as tolerant to defects as lead halide perovskites—a promising alternative to silicon solar cells. Bismuth oxyiodide is also stable in air for at least 197 days, which is a significant improvement over some lead halide perovskite compounds. By sandwiching the bismuth oxyiodide light absorber between two oxide electrodes, they were able to demonstrate a record performance, with the device converting 80 per cent of light into electrical charge.

Bismuth-based devices can be made using common industrial techniques, suggesting that they can be produced at scale and at low cost.